MODULE 2: ECONOMICS OF DEFI ·*SESSION6*

Session 6: Stablecoin Risks & Terra/Luna Case Study

Deconstructing Algorithmic Failures, Reflexivity, and Systemic Contagion

Sudhanshu Pani

Why the Terra/Luna Collapse Matters

$40B

Wiped out in 6 days across UST & LUNA.

75%

Of total UST supply deposited in Anchor.

20%

Subsidized APY promised to depositors.

Core Takeaway

Unlike custodial or collateralized depegs, Terra proved that an algorithmic stablecoin can remain *technically functional* on-chain while its underlying economic arbitrage mechanism experiences total collapse.

Unbacked Algorithmic Mint/Burn Mechanics

Minting UST ($1 Expansion)

Burn $1 LUNA → Mint 1 UST

Demand for UST reduces circulating LUNA supply, pushing LUNA price up during expansion phases.

Redeeming UST ($1 Contraction)

Burn 1 UST → Mint $1 LUNA

To exit UST, users burn UST for newly minted LUNA. Arbitrageurs profit when UST trades below $1.00.

The Structural Flaw

The system inherently assumed LUNA's market capitalization would always remain significantly larger than UST's debt liability, allowing it to absorb redemptions without hyperinflating.

Anchor Protocol: The Unaffordable Yield Machine

Subsidized 20% Yield: Anchor offered $\sim 20\%$ fixed APY on UST, far exceeding organic market lending rates.

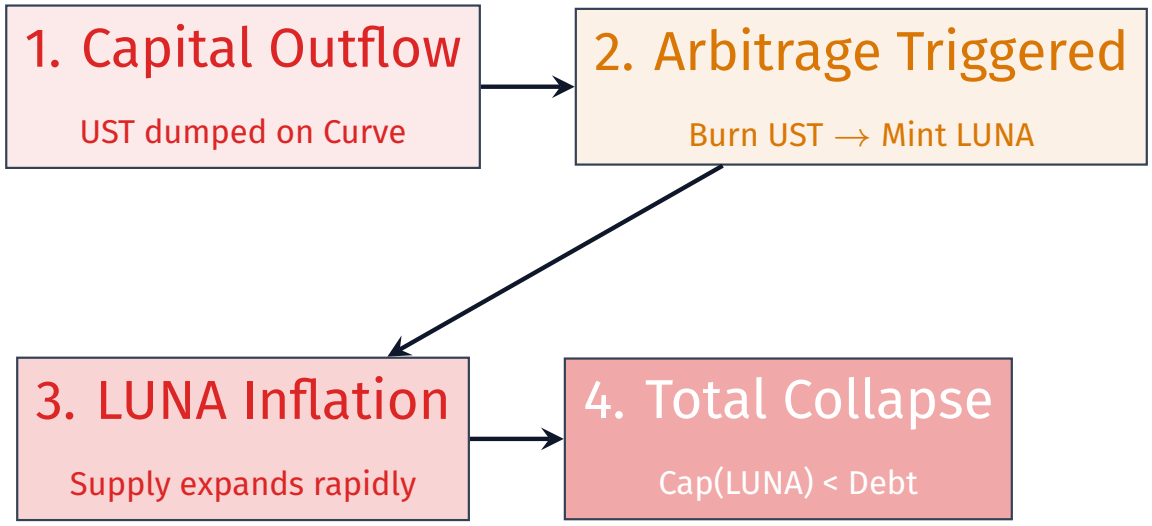
Extreme Concentration: Over $14B of the $18B total UST supply ($\sim 75\%$) was locked inside Anchor alone.

Reserve Depletion: Organic yield was under 6%. The 14% gap was artificially funded by LFG treasury injections.

The Yield Trap

UST demand was not organic; it was driven almost entirely by yield speculation. When the subsidy became unsustainable, capital outflow was inevitable.

Anatomy of the Death Spiral



The Reflexive Loop

As LUNA's price fell, the contract had to mint exponentially *more* LUNA tokens to fulfill every $1 UST redemption. This hyperinflation drove LUNA's price to zero in days.

Chronology of the Collapse (May 2022)

May 07 **Initial Withdrawals:** $400M UST pulled from Anchor; $85M swapped for USDC on Curve. UST dips to $0.985.

May 09 **Peg Cracks:** LFG deploys $750M BTC reserves to defend the peg. Anchor deposits drop to $9B. UST falls to $0.35.

May 11 **Hyperinflation:** LUNA circulating supply skyrockets from 340M to 6.5 Trillion tokens as UST holders panic redeem.

May 13 **Terminal Failure:** Terra blockchain halted twice. Major CEXs delist LUNA/UST. Total ecosystem loss: $\sim \$40$B.

Domino Effect: The Crypto Credit Crisis

3 Arrows Capital

$462M LUNA position collapsed. Inability to cover margin calls led to default and fund liquidation.

Celsius Network

Heavily exposed to yield strategies; froze deposits in June 2022 before declaring bankruptcy.

Voyager Digital

Faced shutdown due to massive uncollateralized loan exposure to 3AC.

Transmission Mechanism

Uncollateralized, interconnected leverage. When the foundational stablecoin asset collapsed, credit was recalled across the entire centralized-decentralized ecosystem.

Impact on Indian Investors & Tax Regime

Speculative Retail Catching

- WazirX reported over 160,000 Indian holders during the depeg event.
- Holder counts rose 77% during the crash as retail investors tried to “catch the falling knife.”

VDA Tax Implications (30% Rule)

- Under Indian VDA tax rules, investors **cannot offset losses** from UST/LUNA against gains in other assets.
- Created severe tax liabilities despite net portfolio destruction.

Counterfactual: Over-Collateralization Comparison

Resilience Benefits

- External collateral (e.g., ETH) prevents death spirals.
- Bank runs trigger orderly liquidations rather than native token hyperinflation.

Capital Inefficiency

- Requiring 150% collateral would have capped UST supply at a fraction of its $18B peak.
- Would have rendered the 20% Anchor yield model impossible to scale.

Takeaway: Over-collateralization saves the peg, but eliminates hyper-growth capabilities.

Modern Evolution: Synthetic USDe vs Terra UST

Dimension	Terra UST (Failed)	Ethena USDe (Synthetic)
Backing Asset	Unbacked (LUNA mint/burn)	stETH + Short ETH Futures
Yield Engine	LFG Treasury Subsidies	Staking Rewards + Funding Rates
Primary Risk	Endogenous Death Spiral	Negative Funding Rates / CEX Risk

Risk Management Lessons for Protocols

Yield is Risk: Unsustainable high yield ($\sim 20\%$) is a structural vulnerability indicator, not organic market adoption.

Avoid Endogenous Collateral: Never back a stablecoin with an asset whose value depends entirely on that stablecoin's adoption.

Stress Test Contagion: Measure cross-protocol leverage and counterparty exposures prior to treasury asset allocations.

IN-CLASS DEBATE

“Can unbacked algorithmic stablecoins ever survive with advanced market making, or is exogenous collateral an absolute necessity?”

Pro-Algorithmic Side:
Scalability requires capital efficiency. Dynamic interest rate controls and automated open-market operations can defend pegs.

Pro-Collateral Side:
Endogenous backing is economically flawed. Without independent, exogenous collateral reserves, death spirals are mathematically inevitable.